
AR 509: Deep Learning for Robotics

Course Code	AR 509
Course Name	Deep Learning for Robotics
L-T-P-C	3-0-2-4
Prerequisites	Consent of faculty advisor
Intended for	PG and PhD
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Deep Learning for Robotics:** Supervised learning for robotics applications, Backpropagation to train neural networks, Overfitting, and Neural network architecture for several robot functions. (6 hours)
- **Neural networks for robot motion control:** Neural networks for inverse kinematic motion calculation, Training with techniques such as dropout and regularization, Solving high-dimensional problems by dimension reduction with principal component analysis (PCA). (10 hours)
- **Reinforcement Learning:** Write a reinforcement learning agent with PyTorch, Overview of Reinforcement Learning Coach - a state-of-the-art reinforcement learning framework. (10 hours)
- **Temporal data and neural networks:** Backpropagation through time and vanishing or exploding gradients, Variations of recurrent neural networks (RNN), and LSTMs to implement them in PyTorch. (10 hours)
- **Laboratory/practical/tutorial Modules:** Neural networks for robot motion control, Reinforcement Learning, Temporal data and neural networks. (6 hours)

Text Books

1. Iosifidis A. and Tefas A., **Deep Learning for Robot Perception and Cognition**, Elsevier.
2. Arana-Daniel N., Alanis A. Y., Lopez-Franco C., **Neural Networks for Robotics: An Engineering Perspective**, CRC Press.
3. Nath V. and Levinson S. E., **Autonomous Robotics and Deep Learning**, Springer.

References

1. Sutton R. and Barto A., **Reinforcement Learning: An Introduction**, MIT Press.
2. Russell S. and Norvig P., **Artificial Intelligence: A Modern Approach**, Prentice Hall.